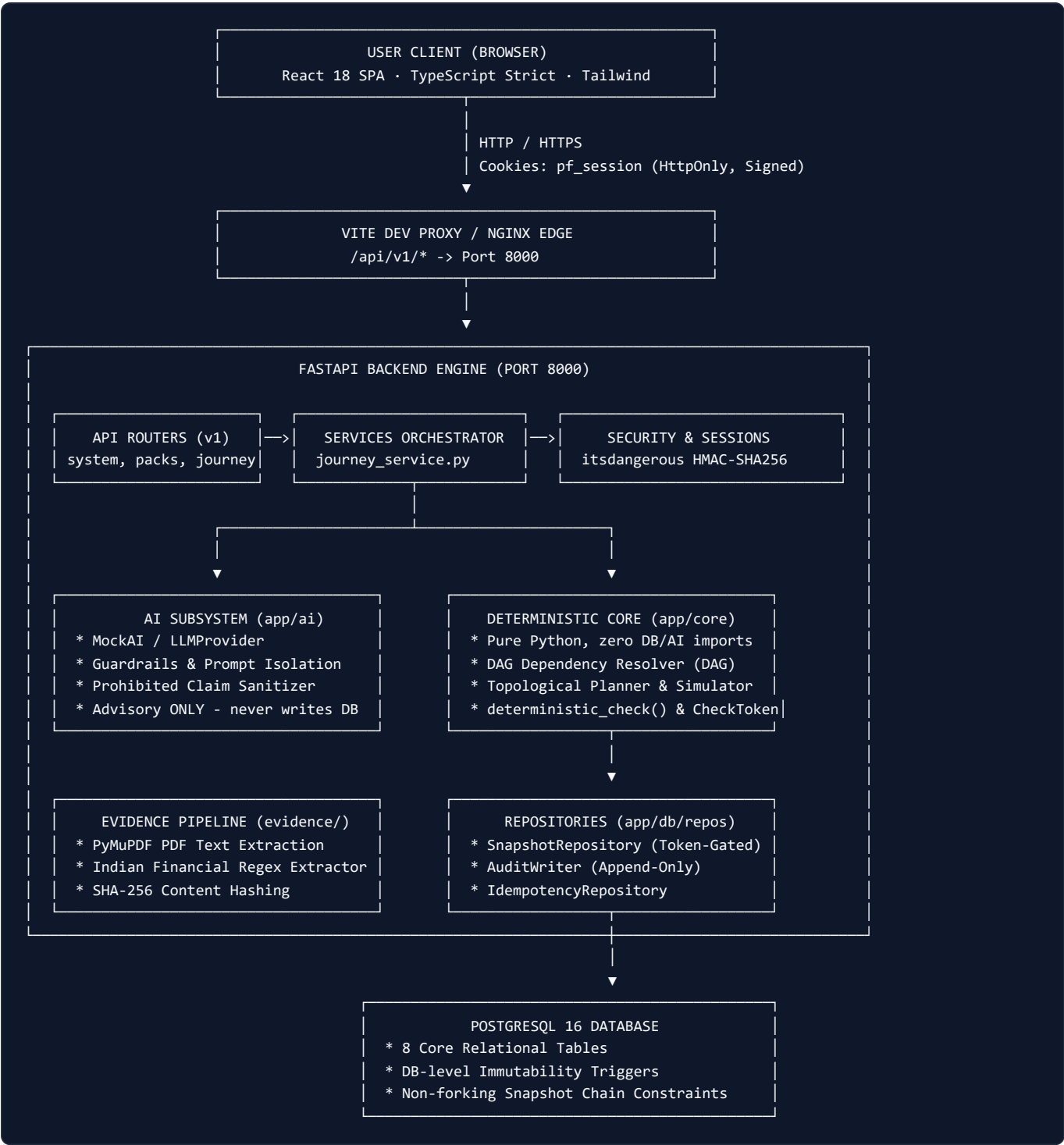


PAYTMFLOW ARCHITECTURAL KNOWLEDGE PACKAGE

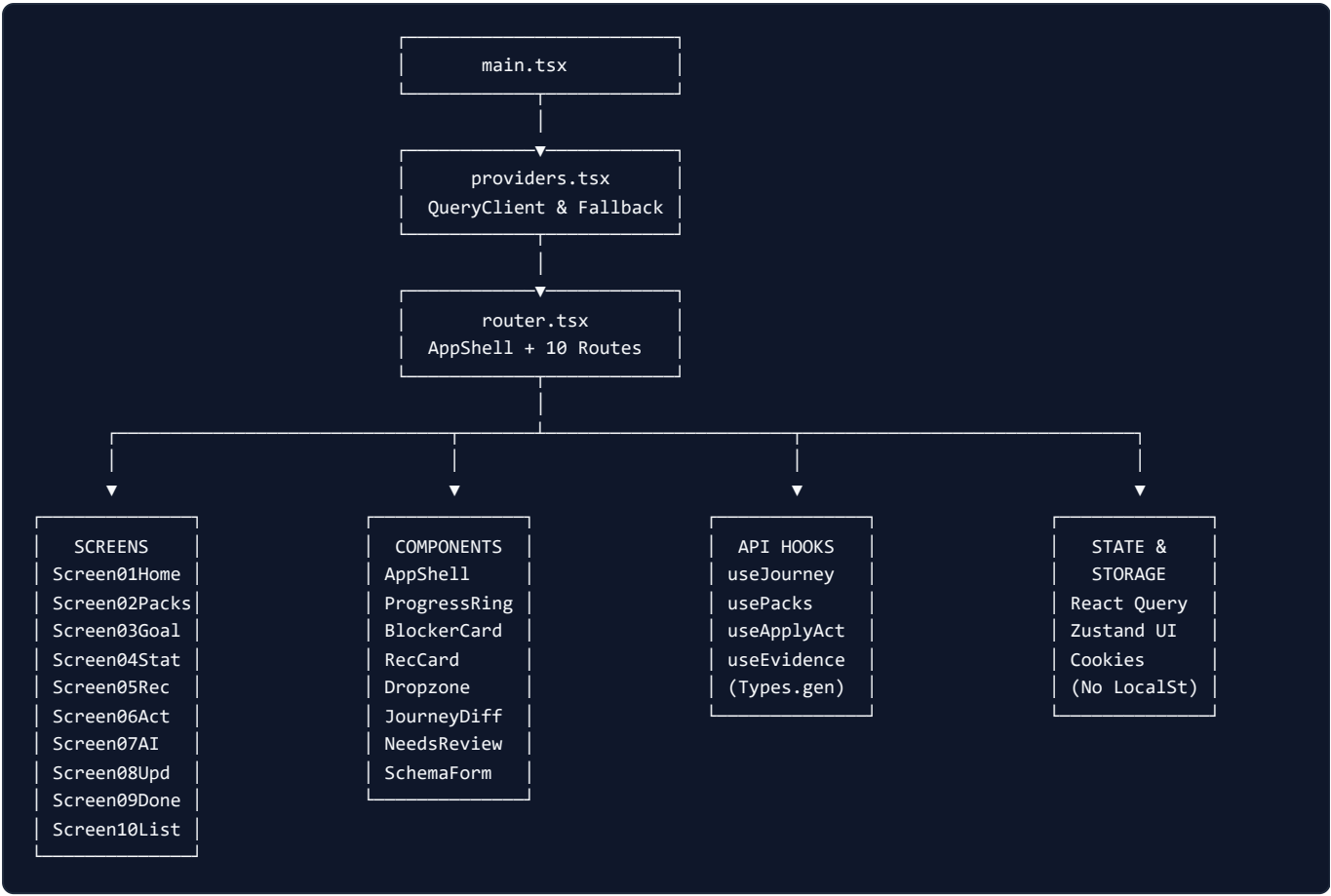
PaytmFlow — Architecture & Technical Reference Manual

Document Type: Developer Quick Reference & System Architecture Specification
Status: Authoritative Repository Audit
Target Audience: Systems Architects, Core Backend Engineers, Frontend Engineers, Tech Leads
Format: Table-Driven, Diagram-Rich, Machine & Human Readable

1. System Architecture Overview



2. Frontend Architecture Diagram



3. Backend Architecture Diagram

```

backend/app/
├─ api/v1/          <-- Presentation (FastAPI REST Routes)
│  └─ system.py      [GET /health, GET /session]
│  └─ packs.py        [GET /journey-packs, GET /journey-packs/{type}]
│  └─ journeys.py     [POST /journeys, GET /journeys/{id}, GET /journeys]
│  └─ recommendation.py [GET /journeys/{id}/recommendation]
│  └─ evidence.py      [POST /journeys/{id}/evidence]
│  └─ actions.py       [POST /journeys/{id}/actions] (THE Mutation Seam)
│  └─ clarifications.py [POST /journeys/{id}/clarifications]
│  └─ diff.py          [GET /journeys/{id}/diff]
├─ services/         <-- Application Orchestration
│  └─ journey_service.py (Coordinates Core + AI + DB)
├─ core/             <-- Pure Deterministic Business Logic (NO I/O)
│  └─ models.py        (CoreSnapshot, CoreFieldState, CheckToken)
│  └─ dependencies.py  (DAG Validation & Topological Sort)
│  └─ rules.py          (Fixpoint Status Resolution)
│  └─ simulate.py       (Pure Consequence Simulator)
│  └─ planner.py        (Shortest Recovery Path Engine)
│  └─ readiness.py      (Deterministic Readiness Evaluator)
│  └─ diff.py           (State Diff Engine)
│  └─ deterministic_check.py (The Mutation Choke Point)
├─ ai/               <-- Isolated Advisory Layer
│  └─ provider.py       (AIProvider Protocol)
│  └─ mock.py           (MockAI - Default High-Fidelity Engine)
│  └─ llm.py            (OpenAI-Compatible LLM Adapter)
│  └─ guardrails.py     (Injection Boundary & Banned Claim Filter)
├─ evidence/         <-- Document Processing Pipeline
│  └─ extract.py        (PyMuPDF Text & Regex Financial Parser)
│  └─ reconcile.py      (Target Mapping & Conflict Heuristics)
│  └─ storage.py         (SHA-256 Hashed File Storage)
└─ db/               <-- Persistence Layer
   └─ models.py         (SQLAlchemy 2 Declarative Models)
   └─ session.py        (Async Session Maker)
   └─ repositories/     (Snapshots, Journeys, Evidence, Audit, Idempotency)

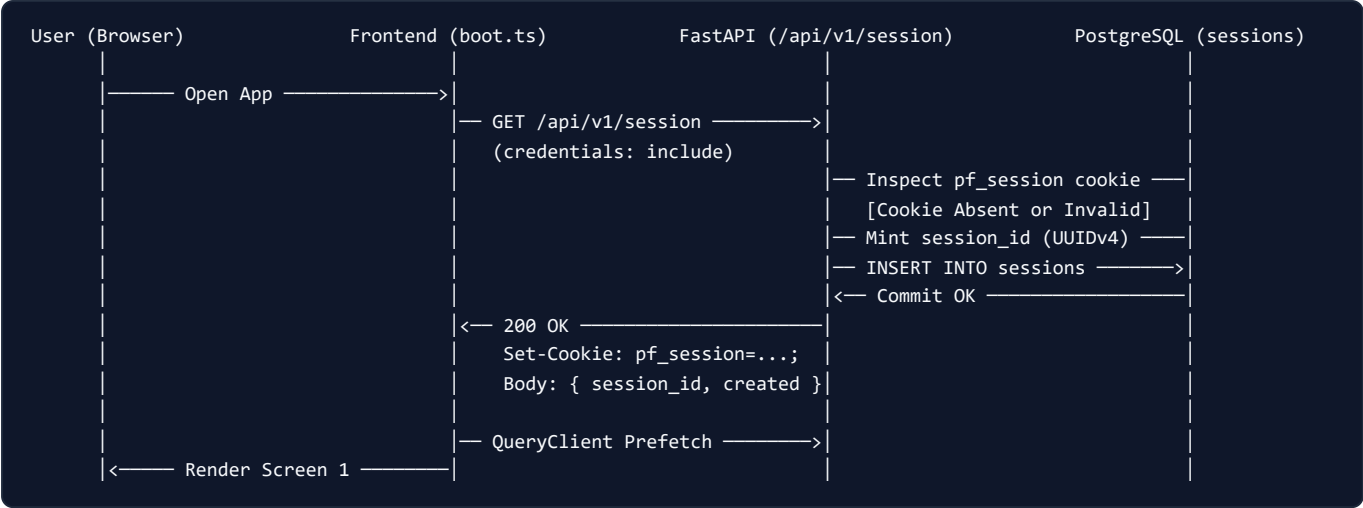
```

4. Database Schema & Entity Relationships



5. End-to-End Sequence Diagrams

A. App Boot & Session Minting Sequence



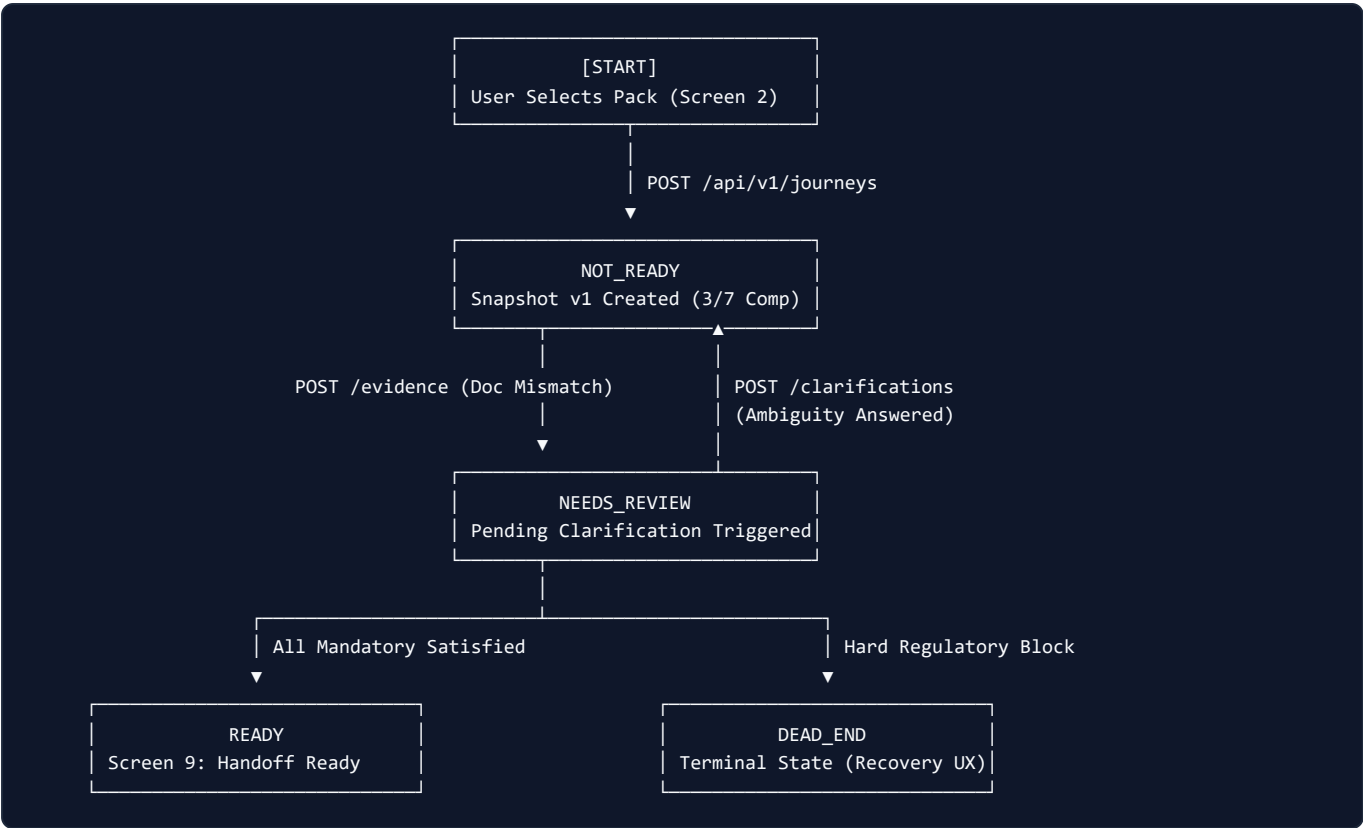
B. Evidence Upload & Preview Sequence (Zero Mutation)



C. State Mutation & Snapshot Transition Sequence



6. Journey State Machine



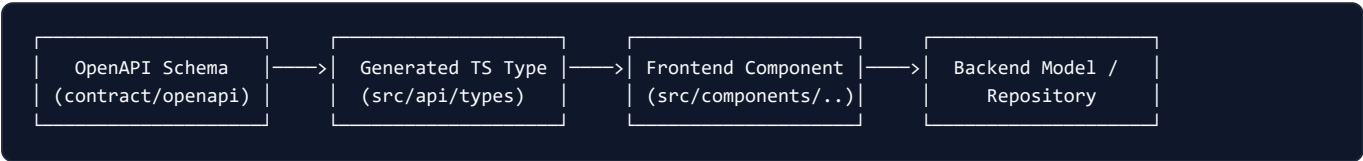
7. Route Map Table

Route	Screen Name	Component	Primary Hook / Query	Key Parameters
/	Home & Landing	Screen01Home	useJourneyList()	None
/start	Journey Selection	Screen02JourneySelection	usePacks()	None
/start/:type	Goal & Basic Info	Screen03GoalBasicInfo	usePack(type) , useCreateJourney()	type (JourneyType)
/j/:id	Current Status	Screen04CurrentStatus	useJourney(id)	id (Journey UUID)
/j/:id/next	Recommendation	Screen05Recommendation	useRecommendation(id)	id (Journey UUID)
/j/:id/act/:actionId	Provide Input / Evidence	Screen06UploadEvidence	useUploadEvidence() , useApplyAction()	id , actionId
/j/:id/analysis	AI Analysis & Preview	Screen07AiAnalysis	Route State (EvidenceResponse)	id (Journey UUID)
/j/:id/updated	Updated Status & Diff	Screen08UpdatedStatus	Route State (ActionResponse)	id (Journey UUID)
/j/:id/complete	Complete Journey	Screen09CompleteJourney	useJourney(id)	id (Journey UUID)
/my-journeys	My Journeys Dashboard	Screen10MyJourneys	useJourneyList()	Tab query param

8. API Endpoints Table

Method	Path	OperationId	Mutates?	Auth / Sec
GET	/api/v1/health	getHealth	No	None
GET	/api/v1/session	getSession	No (Issues cookie)	Cookie / HttpOnly
GET	/api/v1/journey-packs	listJourneyPacks	No	sessionCookie
GET	/api/v1/journey-packs/{journey_type}	getJourneyPack	No	sessionCookie
GET	/api/v1/journeys	listJourneys	No	sessionCookie
POST	/api/v1/journeys	createJourney	Yes (Creates v1)	sessionCookie
GET	/api/v1/journeys/{journey_id}	getJourney	No	sessionCookie
GET	/api/v1/journeys/{journey_id}/recommendation	getRecommendation	No	sessionCookie
POST	/api/v1/journeys/{journey_id}/evidence	uploadEvidence	No (Preview Only)	sessionCookie
POST	/api/v1/journeys/{journey_id}/actions	applyAction	Yes (Sole Mutation)	sessionCookie
POST	/api/v1/journeys/{journey_id}/clarifications	submitClarification	Yes (Disambiguates)	sessionCookie
GET	/api/v1/journeys/{journey_id}/diff	getDiff	No	sessionCookie
POST	/api/v1/demo/reset	resetDemo	Yes (Wipes DB)	X-Demo-Secret

9. Contract-to-Implementation Traceability



Canonical Contract Schema	Generated TypeScript Interface	Backend Core / Pydantic Model	Frontend Component
JourneyStateResponse	components['schemas']['JourneyStateResponse']	app.schemas.journeys.JourneyStateResponse	Screen0: useJour
JourneyPackDetail	components['schemas']['JourneyPackDetail']	app.schemas.packs.JourneyPackDetail	Screen0: SchemaFo
ActionOption	components['schemas']['ActionOption']	app.schemas.journeys.ActionOption	Recommen ActionL:
SimulationPreview	components['schemas']['SimulationPreview']	app.core.models.CoreSimulationResult	Screen0: Consequ
JourneyDiff	components['schemas']['JourneyDiff']	app.core.models.CoreJourneyDiff	JourneyI 7 & Screer
EvidenceResponse	components['schemas']['EvidenceResponse']	app.schemas.evidence.EvidenceResponse	Screen0: Screen0:
FieldState	components['schemas']['FieldState']	app.core.models.CoreFieldState	Blocker: StatusB:
ProgressCounts	components['schemas']['ProgressCounts']	app.core.models.CoreProgressCounts	Progres:
Readiness (Enum)	components['schemas']['Readiness']	app.core.models.CoreReadiness	StatusB: AppShel

10. Complete Error Matrix

HTTP Status	Error Code	Error Message Definition	Root Cause / Trigger	Frontend Mapping & Reaction
400	VALIDATION_ERROR	"Required input field 'X' is missing"	Input does not conform to <code>GoalFieldSpec</code> or number out of bounds	Inline field errors rendered on <code><SchemaForm /></code> ; retains typed inputs.
400	INVALID_JOURNEY_TYPE	"Journey type 'XYZ' is not supported"	Path parameter does not match registered 6 packs	Displays <code>ErrorState</code> card and routes user back to Screen 2.
404	NOT_FOUND	"Journey not found"	Unknown journey ID or cross-session access attempt	"This journey is not available"; routes user to <code>/my-journeys</code> .
409	ACTION_STALE	"The journey state has changed since this action was requested"	<code>expected_snapshot_id</code> $\neq$ <code>current_snapshot_id</code>	Amber banner shown; queries invalidated; no blind action retry .
413	PAYLOAD_TOO_LARGE	"That file is over 10MB"	Uploaded file size exceeds <code>10,485,760</code> bytes	Dropzone remains active; inline message shown; prevents network transmission.
422	ACTION_INVALID	"Action precondition 'X' is not satisfied"	Action invoked out of topological order or unlisted	Disables action CTA; re-fetches recommendation; displays alert toast.
500	INTERNAL_ERROR	"An unexpected error occurred"	Unhandled server exception	Exponential backoff retry once; displays <code><ErrorState /></code> with Retry button.

11. Testing & Validation Matrix

Test Domain	Framework	File Path	Tests Count	Validation Scope
Contract Schemas	pytest	backend/tests/contract/test_openapi_matches.py	2	Asserts FastAPI api generated OpenAPI matches committed openapi.yaml.
Fixture Integrity	pytest	backend/tests/contract/test_fixtures_match_schema.py	36	Validates every JS fixture file against JSON schemas.
Deterministic Invariants	pytest	backend/tests/unit/test_invariants.py	27	Proves 100 runs of simulation produce identical results; r core.
DAG & Topo Planner	pytest	backend/tests/unit/test_planner.py	6	Tests topological and cycle rejection all manifests.
Prompt Injection	pytest	backend/tests/safety/test_prompt_injection.py	15	Verifies <untrusted_domain> boundary tags iso adversarial prompt
Scenario Runner	pytest	backend/tests/scenarios/test_scenarios.py	37	Runs 6 scenarios across all 6 journeys
Frontend Primitives	Vitest	frontend/src/components/primitives/primitives.test.tsx	12	Tests Button, Input, MoneyInput, Badge, Modal, and Tabs focusability.
Dynamic Schema Form	Vitest	frontend/src/components/SchemaForm/SchemaForm.test.tsx	18	Validates runtime compilation from schema specs.
Claim Safety Scan	Vitest	frontend/tests/unit/claim-safety.test.tsx	10	Scans all rendered strings for banned (approved , guaranteed).
Idempotency Client	Vitest	frontend/tests/unit/idempotency.test.tsx	8	Asserts single UUID generated per click duplicate clicks re UUID.
E2E Golden Path	Playwright	frontend/tests/e2e/golden-path.spec.ts	6	Complete headless browser walk through Screens 1 to 9 against MSW.

12. Environment Configuration Reference

Backend (`backend/.env`)

```
DATABASE_URL=postgresql+psycopg://paytmflow:paytmflow@localhost:5432/paytmflow
APP_ENV=local                # local | ci | demo
SESSION_COOKIE_NAME=pf_session
SESSION_SECRET=change-me-32-bytes-minimum-session-secret-key-1234
SESSION_TTL_DAYS=30
CORS_ORIGINS=http://localhost:5173
AI_PROVIDER=mock             # mock | llm (mock is default and demo setting)
AI_TIMEOUT_SECONDS=4
AI_API_KEY=                  # Required only if AI_PROVIDER=llm
AI_MODEL=gpt-4o-mini
EVIDENCE_STORAGE_DIR=./storage/evidence
EVIDENCE_MAX_BYTES=10485760  # 10 MB (Matches UI 10MB copy)
DEMO_RESET_SECRET=change-me-demo-secret
LOG_LEVEL=INFO
```

Frontend (`frontend/.env`)

```
VITE_API_MODE=mock           # mock | live
VITE_API_BASE=/api/v1
VITE_SHOW_DEV_BADGES=true    # Set false for production demo
```

13. Related Documents

- [Complete Project Guide & Master Document](#)
- [Developer & Operations Playbook](#)
- [Shared Development Contract](#)
- [Canonical OpenAPI Specification](#)